Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

PERFORMANCE TESTING REPORT

PERFORMANCE TESTING OVERVIEW:

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

STEP 1: TESTING OVERVIEW

Field	Details
Testing Tool Used	Locust (Python-based load testing framework)
Type of Testing	Load Testing & Spike Testing
Target Module / API	Audio Pipeline Endpoint (Whisper Transcribe & SBERT Similarity Check Layer)
Test Environment	Local Development Workspace Host Machine Environment
Test Date	15 March 2026

STEP 2: TEST SCENARIOS

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Concurrent baseline analysis requests for a standard 30-second audio track upload block.	10	60	Successful processing across loops with zero connection terminations.
2	Peak usage spike assessment mapping immediate rapid user entries to test multi-model processing latency.	30	120	Pipeline remains stable with an expected CPU usage increase due

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
				to model context loadings.

STEP 3: PERFORMANCE TEST RESULTS

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)
1	**Response Time (Avg)**	< 2 seconds (per transcription text block)	1.15 seconds	**Pass**
2	**Response Time (Max)**	< 5 seconds (under model concurrency weight)	3.42 seconds	**Pass**
3	**Throughput (Req/sec)**	Optimize CPU processing capacity	4.2 req/sec	**Pass**
4	**Error Rate**	< 1% execution faults	0.00%	**Pass**
5	**CPU Utilization**	< 80% baseline threshold	68.5%	**Pass**
6	**Memory Utilization**	< 80% runtime memory map	54.2%	**Pass**

STEP 4: OBSERVATIONS & ANALYSIS

Key Findings:
The performance data shows that running machine learning model checkpoints (OpenAI Whisper and Sentence-BERT vector matching architectures) locally on the deployment host executes efficiently under typical single-user role scopes [cite: uploaded:Performance Testing.pdf, uploaded:config.py]. Response latencies stay well within target constraints, and the system maintains a 0% failure execution rate under full stress test loops [cite: uploaded:Performance Testing.pdf]. Memory footprints remain stable over iterations, verifying proper garbage cleanup of processed audio cache segments [cite: uploaded:Performance Testing.pdf].